STAFF REVIEW · H446 287 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Original can be reconstructed exactly.

2. Consecutive repeated instances of the same symbol.

3. Extra metadata needed to represent/decode data.

4. [2,6,4,10], sum 22; indexes are positions 1 and 3.

I Do — source worked steps

1. Predict first — Identify maximal consecutive runs: four B, two A and one C.

2. Model and check — Output each run as value/count and expand them back to verify exact reconstruction.

We Do — guided CHECK

1. Predict first
Use dictionary reference 1 for each repeated word while preserving the space separator.
Model and check
Decoder reads both references with its matching dictionary and reconstructs the original phrase.

You Do — source mark schemes

1. (A,5) [1], (B,2) [1], (C,3) [1], (D,4) [1].

2. 0 [1], space [1], 1 [1], space [1], 0 [1]; decoder with same dictionary recovers exact phrase [1].

3. No long runs [1], counts add overhead [2]; repeated substrings dictionary references [2], dictionary itself costs space [1].

Exit Ticket — source answer

1. False. It may expand files with few repeated runs.